



NeuroSpace

Complete evaluation export

Job ID	a910ece4-a0c1-4e3c-a6b6-ff45278fcc7e
Status	completed
Result timestamp	2026-05-18T11:10:20.761078
PDF generated	2026-05-18 12:15:47

Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	pca
feature_selector	pca
manual_overrides.rnn_epochs	20
manual_overrides.qrnn_epochs	10
manual_overrides.batch_size	32
manual_overrides.rnn_batch_size	32
manual_overrides.qrnn_batch_size	16
num_classes	2
class_names	0, 1
qrnn_backend	qiskit
noise_enabled	False
noise_level	0.0000
mitigation_enabled	False
mitigation_runs	1

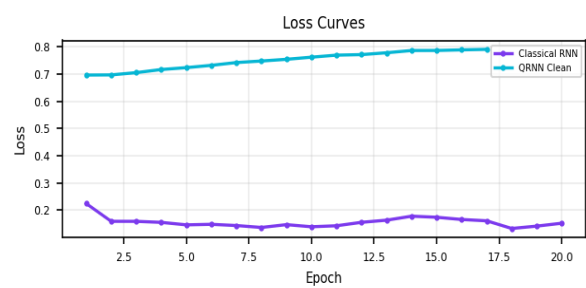
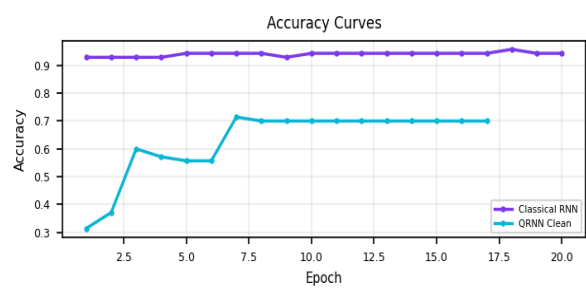
Best Model

Winner	RNN Accuracy	QRNN Accuracy	Improvement
RNN	96.43%	85.00%	11.43%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss
Classical RNN	96.43%	96.57%	96.43%	96.45%	0.1525
QRNN Clean	85.00%	86.33%	85.00%	84.02%	0.7911

Learning Curves



Classical RNN

Metric	Value
Accuracy	96.43%
Precision	96.57%
Recall	96.43%
F1 Score	96.45%
Loss	0.1525
Train Loss	0.0635
Val Loss	0.1525
Train Accuracy	97.96%
Val Accuracy	94.29%
Train Size	489
Val Size	70
Test Size	140

Classical RNN Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	88	4
Class 1	1	47

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
0	98.88%	95.65%	97.24%	92.0000
1	92.16%	97.92%	94.95%	48.0000
macro avg	95.52%	96.78%	96.09%	140.0000
weighted avg	96.57%	96.43%	96.45%	140.0000
Accuracy			96.43%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.2566	0.2250	92.02%	92.86%
2	0.1271	0.1595	96.32%	92.86%
3	0.0935	0.1594	97.14%	92.86%
4	0.0825	0.1557	97.75%	92.86%
5	0.0890	0.1465	97.34%	94.29%
6	0.0858	0.1485	97.55%	94.29%
7	0.0826	0.1440	97.55%	94.29%
8	0.1001	0.1369	97.75%	94.29%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
9	0.1050	0.1473	96.73%	92.86%
10	0.0674	0.1395	97.96%	94.29%
11	0.0712	0.1431	97.55%	94.29%
12	0.0892	0.1559	97.14%	94.29%
13	0.0784	0.1636	98.16%	94.29%
14	0.0624	0.1783	98.36%	94.29%
15	0.0869	0.1744	97.14%	94.29%
16	0.0606	0.1662	97.75%	94.29%
17	0.0517	0.1614	98.16%	94.29%
18	0.0695	0.1332	97.96%	95.71%
19	0.0595	0.1418	97.75%	94.29%
20	0.0635	0.1525	97.96%	94.29%

QRNN Clean

Metric	Value
Accuracy	85.00%
Precision	86.33%
Recall	85.00%
F1 Score	84.02%
Loss	0.7911
Train Loss	0.6061
Val Loss	0.7911
Train Accuracy	85.69%
Val Accuracy	70.00%
Train Size	489
Val Size	70
Test Size	140

QRNN Clean Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	90	2
Class 1	19	29

QRNN Clean Classification Report

Class	Precision	Recall	F1 Score	Support
0	82.57%	97.83%	89.55%	92.0000
1	93.55%	60.42%	73.42%	48.0000
macro avg	88.06%	79.12%	81.48%	140.0000
weighted avg	86.33%	85.00%	84.02%	140.0000
Accuracy			85.00%	

QRNN Clean Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.7167	0.6967	31.29%	31.43%
2	0.6850	0.6973	33.54%	37.14%
3	0.6671	0.7056	66.05%	60.00%
4	0.6542	0.7170	73.01%	57.14%
5	0.6437	0.7240	73.62%	55.71%
6	0.6354	0.7324	76.89%	55.71%
7	0.6299	0.7426	85.48%	71.43%
8	0.6244	0.7483	85.69%	70.00%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
9	0.6219	0.7545	85.48%	70.00%
10	0.6184	0.7623	85.48%	70.00%
11	0.6147	0.7698	85.07%	70.00%
12	0.6135	0.7718	85.89%	70.00%
13	0.6105	0.7786	85.48%	70.00%
14	0.6123	0.7868	84.87%	70.00%
15	0.6058	0.7870	85.89%	70.00%
16	0.6077	0.7892	85.48%	70.00%
17	0.6061	0.7911	85.69%	70.00%